



GATE CS / IT 2027

Computer Networks

Quick Revision Sheet

Complete GATE syllabus coverage — All topics and subtopics

Network Models

OSI Layer	TCP/IP Layer	Key Protocols	PDU
Application (7)	Application	HTTP, FTP, SMTP, DNS, DHCP, Telnet	Message
Presentation (6)	Application	SSL/TLS, JPEG, ASCII	Message
Session (5)	Application	NetBIOS, PPTP	Message
Transport (4)	Transport	TCP, UDP	Segment
Network (3)	Internet	IP, ICMP, ARP, RARP, Router	Packet
Data Link (2)	Network Access	Ethernet, PPP, Switch, Bridge	Frame
Physical (1)	Network Access	Hub, Cables, NIC, Repeater	Bits

IP Addressing & Routing

Classes & Subnetting

- Class A: /8 (0.x.x.x), Class B: /16 (128-191.x.x.x), Class C: /24
- Usable hosts = $2^{(\text{host bits})} - 2$
- Subnet mask: 255.255.255.0 = /24
- CIDR allows non-classful subnetting
- Private: 10.x, 172.16-31.x, 192.168.x

Routing Protocols

- RIP: distance vector, max 15 hops, slow convergence
- OSPF: link state, fast convergence, Dijkstra algorithm
- BGP: path vector, used between ISPs (inter-AS routing)
- Classful routing ignores subnet mask

TCP vs UDP

Feature	TCP	UDP
Connection	Connection-oriented (3-way handshake)	Connectionless
Reliability	Reliable (ACK, retransmit)	Best effort
Ordering	In-order delivery	No ordering guarantee
Flow Control	Sliding window (receiver)	None
Congestion	Slow start, AIMD	None
Overhead	20-60 byte header	8 byte header
Use Cases	HTTP, FTP, SMTP, SSH	DNS, VoIP, Video, Gaming

Error & Flow Control

Protocols

- Stop & Wait: efficiency = $1/(1+2a)$, a = propagation delay/transmission time
- Go-Back-N: window size $W \leq 2^N - 1$, retransmit all from error
- Selective Repeat: window size $W \leq 2^{(N-1)}$, retransmit only lost frame
- CRC error detection: polynomial division remainder = 0 means no error
- Hamming code: corrects 1-bit, detects 2-bit errors